POSE WITH AI – AI Powered Pose Corrector and Analyzing Camera

A PROJECT REPORT

Submitted by

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Introduction :

POSE WITH AI is an intelligent pose recommendation and posture analysis application that helps users capture better photographs using Artificial Intelligence (AI) and Computer Vision. The application analyses a user's body posture in real time through a camera and provides personalized pose suggestions to improve appearance, confidence, and photo quality.

The system uses AI-based pose estimation techniques to detect key body landmarks such as the head, shoulders, arms, and body alignment. Based on this analysis, it compares the user's posture with predefined ideal poses and provides instant feedback, including pose corrections and improvement suggestions. This enables users to adjust their posture before capturing a photo, resulting in more attractive and professional-looking images.

In addition to pose analysis, the application offers features such as live camera preview, AI-generated pose recommendations, pose scoring, countdown timer, image capture, and photo history. It is designed with a simple and user-friendly interface, making it suitable for users of all age groups without requiring any technical knowledge.

The project demonstrates the practical application of Artificial Intelligence, Machine Learning, and Computer Vision in solving real-world problems. It combines modern technologies to create a smart photography assistant that enhances user experience while promoting confidence and creativity. The application can be further extended with advanced features such as personalized pose recommendations, gesture recognition, fashion suggestions, social media integration, and cloud-based photo storage.

Overall, **POSE WITH AI** provides an innovative solution that bridges AI technology with mobile and web photography, making professional-quality posing guidance accessible to everyone.

Objective :

The primary objective of **POSE WITH AI** is to develop an intelligent application that assists users in achieving better body posture and attractive poses while taking photographs using Artificial Intelligence (AI) and Computer Vision.

The specific objectives of the project are:

- To detect and analyse the user's body posture in real time using AI-based pose estimation.
- To provide intelligent pose recommendations based on the user's current posture.
- To compare the user's pose with predefined reference poses and calculate a pose accuracy score.
- To offer instant feedback and corrective suggestions for improving body alignment and posture.
- To enable users to capture high-quality photographs with confidence through AI-assisted guidance.
- To create a simple, interactive, and user-friendly interface that can be used by people of all age groups.
- To maintain a history of captured photos for future reference.
- To demonstrate the practical application of Artificial Intelligence, Machine Learning, and Computer Vision in the field of smart photography.

The overall goal of the project is to enhance the photography experience by providing real-time AI-powered pose guidance, enabling users to capture professional-looking photographs without requiring expert photography skills.

Modules :

The **POSE WITH AI** application is divided into the following functional modules:

1. User Authentication Module

- Allows users to create a new account and log in securely.
- Supports guest mode for quick access without registration.
- Manages user profiles and session information.

2. Live Camera Module

- Activates the device camera for real-time image capture.
- Displays a live camera preview for pose analysis.
- Enables users to capture photos directly from the application.

3. AI Pose Detection Module

- Detects human body key points such as the head, shoulders, elbows, hips, knees, and ankles using AI-based pose estimation.
- Continuously tracks body posture in real time.

4. Pose Recommendation Module

- Compares the user's posture with predefined reference poses.
- Suggests the most suitable pose based on body alignment.
- Recommends standing, sitting, confident, interview, or other supported poses.

5. Pose Analysis and Scoring Module

- Evaluates the user's pose accuracy.

- Calculates a pose score based on body alignment and posture.
- Provides instant visual feedback on overall pose quality.

6. AI Feedback Module

- Displays corrective suggestions to improve the user's posture.
- Highlights body parts that require adjustment.
- Guides the user to achieve a more balanced and natural pose.

7. Photo Capture and Storage Module

- Captures high-quality images after successful pose analysis.
- Saves captured photos to the application for future reference.
- Maintains a photo history for users.

8. User Interface Module

- Provides an intuitive and responsive interface.
- Displays live camera feed, AI suggestions, pose score, and capture controls.
- Ensures a smooth and user-friendly experience across supported devices.

Together, these modules enable the application to provide real-time AI-powered pose guidance, helping users capture better photographs with improved posture and confidence.

System Architecture :

The **POSE WITH AI** system follows a modular architecture in which the user interacts with the application through a camera interface. The captured image or live video frame is processed by the AI pose detection engine, which identifies key body landmarks and analyses the user's posture. Based on the detected body positions, the pose analysis module evaluates the posture and compares it with predefined reference poses stored in the system.

The AI recommendation module then generates suitable pose suggestions, calculates a pose accuracy score, and provides corrective feedback to help the user improve their posture. Once the user achieves the desired pose, the application allows the image to be captured and stored in the photo history. User account information, captured images, and application data are managed through the backend database, ensuring secure storage and easy retrieval.

The overall architecture consists of the following components:

- **User Interface (Frontend):** Provides login, live camera preview, pose display, AI suggestions, and photo capture options.
- **Camera Module:** Captures live video frames for analysis.
- **AI Pose Detection Engine:** Detects body key points using computer vision techniques.
- **Pose Analysis Module:** Compares the detected posture with reference poses and calculates the pose score.
- **AI Recommendation Module:** Generates pose suggestions and corrective feedback.

- **Backend Server:** Processes application requests, manages user authentication, and handles communication between the frontend and database.
- **Database:** Stores user information, captured photos, and pose history.

Implementation :

APP. PY

```
  from flask import Flask, render_template

app = Flask(__name__)

@app.route("/")
def home():
    return render_template("index.html")

if __name__ == "__main__":
    app.run(
        host="0.0.0.0",
        port=5000,
        debug=True
```

```
}
```

INDEX FILE

```
<!DOCTYPE html>
<html lang="en">

<head>

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Pose With AI v4</title>

<link rel="stylesheet"
href="{{ url_for('static', filename='css/style.css') }}">

</head>

<body>

<div class="container">

<h1>📸 Pose With AI v4</h1>

<p class="subtitle">
```

AI Fashion & Posture Coach

</p>

<div class="camera-container">

<video id="video" autoplay playsinline></video>

<canvas id="canvas"></canvas>

</div>

<div class="controls">

<select id="poseType">

<option value="standing">

Standing Posture

</option>

<option value="sitting">

Sitting Posture

</option>

<option value="confident">

Confident Pose

</option>

<option value="interview">

Interview Pose

</option>

<option value="casual">

Fashion Casual

</option>

<option value="professional">

Fashion Professional

</option>

<option value="traditional">

Fashion Traditional

</option>

<option value="trendy">

Fashion Trendy

</option>

<option value="social">

Social Media Pose

</option>

</select>

```
<button id="startBtn">
Start Camera
</button>

<button id="captureBtn">
Capture Photo
</button>

</div>

<div id="countdown">
</div>

<div class="score-panel">

<h2 id="score">
Posture Score: 0%
</h2>

<h3 id="bestScore">
Best Score: 0%
</h3>

</div>
```

<div class="panel">

<h2>

🤖 AI Recommended Pose

</h2>

<div id="recommendedPose">

Select a mode...

</div>

</div>

<div class="panel">

<h2>

🛠️ AI Corrections

</h2>

<div id="tips">

Waiting...

</div>

```html
</div>

<div class="panel">

<h2>
📷 Pose History
</h2>

<ul id="historyList">

</ul>

</div>

</div>

<script src="https://cdn.jsdelivr.net/npm/@mediapipe/pose/pose.js"></script>

<script
src="https://cdn.jsdelivr.net/npm/@mediapipe/camera_utils/camera_utils.js"></script>

<script src="{{ url_for('static', filename='js/aiSuggestions.js') }}"></script>

<script src="{{ url_for('static', filename='js/poseHistory.js') }}"></script>

<script src="{{ url_for('static', filename='js/capture.js') }}"></script>
```

```
<script src="{{ url_for('static', filename='js/pose.js') }}"></script>

</body>

</html>
```

OUTPUT :-